scReg-Eval: a fixed-panel audit of regulatory alignment in single-cell RNA foundation-model gene graphs

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ABSTRACT

Background. Claims that single-cell RNA foundation models encode gene regulation are difficult to audit because many reference networks share co-expression structure with the models' training data.
Methods. We introduce **scReg-Eval**, a fixed-panel audit protocol and software capsule that compares frozen pairwise foundation-model gene graphs with a sequence- and accessibility-derived regulatory-potential proxy while controlling co-expression, target peak count, gene length, GC content, RNA detection rate, transcription-factor out-degree, and target in-degree. The protocol-frozen panel contains 1,200 genes, including 446 in-range transcription factors (TFs). We evaluate thirteen pooled model/readout combinations across brain and paired peripheral blood mononuclear cell (PBMC) multiome data using two plus-one Monte Carlo randomization tests ($N = 999$ each): a gene-label permutation and a within-TF-row, degree-preserving target shuffle. Benjamini–Hochberg correction is applied within eight prespecified tissue-by-confound-specification-by-null families. The estimand is fixed-panel unusualness of alignment under named confounds and two named randomizations—not causal TF–target recovery and not model ranking.
Results. Dual-null "Support" under the primary full confound specification is seven of thirteen rows (six small positive alignments, ρ from 0.00425 to 0.01241, plus one negative Geneformer attention row in brain), but that count is not regulatory recovery: the same seven rows appear only when degree covariates computed from the proxy are conditioned on; under the co-primary non-degree specification dual-null Support is $0/13$ and three of the six positives change sign; and the six positive dual-null rows fail the protocol's own FM–co-expression concordance check (Spearman -0.18 to $+0.01$), so dual-null Support marks unusual residual geometry under disclosed degree hazard, not recovered regulation. The co-expression baseline, tested on the same edge set, is itself dual-null supported in both tissues, and the PBMC Geneformer-embedding alignment ($\rho = 0.00425$) is smaller than its own baseline ($\rho = 0.0070$). A supervised TF-disjoint probe on paired PBMC data gives mixed family-level results: UCE versus co-expression is supported by paired sign-flip testing ($q = 0.0325$), as is the random-init floor ($q = 0.0300$), whereas Geneformer embedding ($q = 0.5125$), Geneformer attention ($q = 0.65$), and scGPT ($q = 0.255$) are not. The probe therefore must not be summarized as showing that all models are indistinguishable from co-expression.
Conclusions. The audit finds weak, readout-specific, specification-dependent residual alignment concentrated in embedding-geometry readouts. It neither supports a global regulatory-capability claim for these scFM gene graphs nor licenses a field-level negative conclusion beyond this fixed-panel protocol instance.

Keywords: single-cell RNA-seq, foundation models, gene regulatory networks, co-expression confounding, randomization inference, benchmark audit

1 INTRODUCTION

Single-cell RNA foundation models (scFMs) are increasingly used as frozen representations for network inference and perturbation analysis (Theodoris et al., 2023; Cui et al., 2024; Kalfon et al., 2025). A high score against a curated or transcriptome-derived regulatory network, however, need not imply that a model encodes regulation beyond co-expression (Pratapa et al., 2020; Marbach et al., 2012; Nourisa et al., 2025). Both the reference and the model can inherit gene abundance, detectability, hubness, and co-variation, so agreement with a co-expression-laden reference can look like regulatory recovery when it is mostly shared structure. Standardized scFM task suites and living GRN benchmarks have begun to systematize evaluation (Qiu et al., 2025; Wu et al., 2025; Nourisa et al., 2025), but most still score expression-derived or literature GRNs rather than separating edge weights from expression statistics in the reference construction. Without that separation, it is hard to tell whether an scFM gene graph carries residual regulatory alignment after co-expression and simple graph-degree structure are controlled.

We address this problem as an audit-design question rather than as a model-ranking exercise. scReg-Eval freezes a transcription-factor (TF)–target panel, constructs an accessibility-and-motif regulatory-potential proxy whose *edge weights* are assembled without expression statistics, applies the same edge mask and confounds to every model, and evaluates the observed alignment against two structurally different randomization nulls. Panel membership and ATAC peak presence remain expression-adjacent design choices, so the proxy is not claimed to be RNA-structure-independent in every upstream decision—only that its scored edges are not co-expression weights. This design does not make the proxy causal ground truth. It asks a narrower, reproducible question: on the fixed panel, does a model gene graph align with the proxy after controlling measured RNA and graph structure, and is that alignment unusual under both label and target-assignment randomizations? Negative benchmarks of foundation-model interpretability under careful controls have precedent (Ahlmann-Eltze et al., 2025; Boiarsky et al., 2023; Kedzierska et al., 2025); our aim is the audit protocol, software capsule, and reporting discipline rather than a global verdict that scFMs do or do not encode regulation.

The closest prior results show that foundation-model attention recovers co-expression rather than unique regulatory signal (Kendiukhov, 2026b), and that residual-stream features of Geneformer and scGPT likewise encode organized pathway and interaction knowledge with minimal causal regulatory logic (Kendiukhov, 2026a). Our reference differs from transcriptome-derived networks used in those interpretability studies: the proxy edge weights here are built from genomic sequence (motif matches) and chromatin accessibility only, so shared expression statistics alone cannot satisfy the reference through co-expression edges. That construction choice is what makes the audit informative when alignments are small: residual agreement cannot be explained by shared expression edge weights alone, but it also cannot be read as causal TF–target truth. Within one model, we additionally find that two readouts of the same frozen network disagree in sign (Geneformer embedding $+0.0044$ versus attention -0.0036 in brain), a per-readout inconsistency that a model-level regulatory account does not predict and that single-readout studies cannot see.

Our contributions are:

1. a reusable fixed-panel audit protocol and hash-pinned software capsule (panel manifests, null semantics, FDR families, and one-command validation) for scFM gene-graph claims, with biology serving as the application scene rather than as a claim of recovered regulatory truth;
2. a sequence/accessibility regulatory-potential proxy and 446-TF by 1,200-gene panel that show moderate observed agreement across brain, blood, and a mixed fibroblast reprogramming dataset;
3. a full-confound partial-Spearman statistic that controls co-expression and six structural covariates, with a non-degree specification reported as a co-primary sensitivity analysis;
4. two finite-panel Monte Carlo tests with explicit null semantics, deterministic seed streams, plus-one correction, and eight prespecified FDR families;
5. a multi-readout application of Geneformer, scGPT, scFoundation, and UCE (Theodoris et al., 2023; Cui et al., 2024; Hao et al., 2024; Rosen et al., 2026; Yang et al., 2022)—purely transcriptomic FMs, as distinct from knowledge-informed models that inject prior GRN structure at pretraining (Yang et al., 2024; Hu et al., 2026)—showing small, readout-specific primary alignments—including a sign-inconsistent embedding/attention pair within one model—rather than a uniform positive or negative result; and
6. descriptive per-cell-type, cross-tissue, and axis-aligned injection diagnostics whose scope is kept separate from the pooled inferential families.

These choices are reported as ordinary design rules rather than as a separate boxed checklist. We freeze and hash the TF–target panel; label the reference as a regulatory-potential proxy rather than causal truth; control co-expression and structural covariates; state exactly what each randomization changes; use plus-one Monte Carlo p-values; define FDR families before reading results; and keep exploratory cell-type and sensitivity analyses distinct from the primary test. Methods define the panel, proxy, confounds, and nulls; Results report the primary full-confound audit and the co-primary non-degree sensitivity; Discussion and the final figure set bound what the alignments can and cannot claim.

2 METHODS

2.1 Estimand

The target of inference is *fixed-panel unusualness*: on one protocol-frozen TF–target panel, is the partial-Spearman alignment between a frozen foundation-model gene graph and the accessibility/motif proxy unusually large (in absolute value) under (i) a named full or non-degree confound design and (ii) two named randomizations (gene-label permutation and within-TF-row degree-preserving target shuffle), after within-family BH control? Dual-null "Support" answers that panel-local question only. It is not an estimand for causal TF–target effects, not a ranking of foundation models, and not a test of residual regulatory content after every possible confound. Concordance checks against co-expression further bound whether a dual-null row resembles attention-like co-expression geometry; they do not adjudicate residual regulatory truth. Discussion returns to this scope when interpreting Support counts.

2.2 Regulatory-potential proxy and fixed panel

The panel contains 1,200 genes, including 446 unique in-range transcription factors, frozen by a SHA-256 manifest. Accessible peaks from brain snATAC-seq, paired PBMC multiome, and a chemical reprogramming ATAC dataset (GSE206767, in which day-0 BJ fibroblasts are pooled with day-3 induced neuron-like cells as a single `ALL` accessibility track) are linked to genes by a ± 2 kb promoter-plus-gene-body rule, which retains 6,609 of 111,743 accessible peaks in the PBMC data (5.91%) and systematically excludes distal enhancer elements. JASPAR2024 CORE vertebrate motifs are scanned with MOODS (Rauluseviciute et al., 2024; Korhonen et al., 2009); motif evidence in an accessible peak contributes a directed TF–target regulatory-potential weight, following the motif-in-accessible-region linkage paradigm used by regulon-inference methods (Aibar et al., 2017). Edge weights are assembled without expression statistics; panel membership and ATAC peak-presence filters remain expression-adjacent inputs, so the proxy is not claimed RNA-structure-independent in every design choice. Accessibility, motif occurrence, and proximity still do not establish direct regulation or causality. We therefore call it a proxy throughout. The two evaluation tissues differ in pairing: the PBMC data are a paired multiome assay (RNA and ATAC from the same cells), whereas the brain analysis combines snATAC-seq with cross-study RNA, so the co-expression control is intrinsically stronger in PBMC than in brain.

Brain analyses apply a prespecified neural marker mask to every brain readout; in this panel the mask removes 446 of 534,754 brain edges (0.08%), a small fixed-panel exclusion because only one of the 34 prespecified markers (VCAN, which is not a transcription factor) falls in the fixed panel. PBMC edges are unmasked under the prespecified tissue policy. Before analysis, the driver verifies the manifest, TF identity and order across all caches, matrix dimensions and finiteness, and four pinned legacy-file hashes.

2.3 Foundation-model and baseline graphs

Gene graphs are read from cached, model-validated outputs over the same manifest. Available pooled readouts are Geneformer embedding, attention, and in-silico knockout; scFoundation encoder; UCE encoder (4-layer variant, checkpoint SHA-256 pinned by the pipeline); and scGPT encoder; plus a random-init Geneformer floor. Coverage differs by tissue because only precomputed, validated graphs are included: brain has eight FM/readout rows and PBMC has five. Co-expression is reported as a baseline on the same edge set and nulls, but is not included as an FM row or in FM multiplicity families. Co-expression enters its own test only as the signal, never as a covariate: its design matrix is the full specification minus the co-expression column—an intercept plus the six structural covariates (target peak count, gene length, RNA detection rate, GC content, TF out-degree, target in-degree)—because partialling co-expression out of itself is degenerate by construction. CellPLM is outside scope because its architecture does not expose contextual per-gene states (Wen et al., 2024). Inference settings differ by model and are reported rather than harmonized: PBMC readouts use a deterministic 4,000-cell pool for every model;

brain Geneformer embedding and scGPT use all available cells, brain Geneformer attention is capped at 4,000 cells, brain UCE uses 2,000 cells, and brain scFoundation uses 1,500 cells with a 512-token context cap. Manifest gene coverage is 1,159/1,200 for PBMC UCE, 1,195/1,200 for brain UCE, 1,199/1,200 for brain scFoundation, 1,168/1,200 for PBMC scFoundation, and 1,200/1,200 otherwise.

2.4 Observed statistic and confound specifications

For each retained TF–target pair, all graph weights are rank transformed. The primary statistic is the Pearson correlation between residualized ranks, equivalent to partial Spearman correlation. The *full* design contains an intercept, ranked co-expression, and standardized target peak count, gene length, RNA detection rate, GC content, TF out-degree, and target in-degree. Peak-count and GC covariates are computed from at most 40 peaks per gene, so they understate the extremes of very peak-dense genes. The two degree covariates are computed from the proxy graph itself, so the full specification conditions on functions of the outcome; this is a validity hazard, not a bookkeeping detail, and it is why the *non-degree* design, which omits both degree columns, is reported co-primary rather than as a footnote. Observed and randomized statistics always use matching column sets.

2.5 Finite-panel randomization inference

The TF panel is treated as fixed; no TF-superpopulation bootstrap (Efron and Tibshirani, 1993) or confidence interval is used. For the gene-label null, one permutation relabels both endpoints of the proxy graph, in the spirit of the Mantel matrix correspondence test (Mantel, 1967). The FM graph, co-expression, and non-degree gene covariates remain fixed. In the full specification, proxy-derived TF out-degree and target in-degree are recomputed after relabeling.

The second null independently shuffles edge weights within each TF row across non-self targets. This preserves TF out-degree and the row's weight multiset while changing target assignment; target in-degree is recomputed only for the full specification. One proxy randomization per replicate is shared across model rows, with each model scored against the same replicate. Explicit-index tests verify numerical equivalence to separate least-squares calculations, including rank-deficient designs.

Both tests use two-sided plus-one correction (Phipson and Smyth, 2010),

$$p_{\mathrm{MC}} = \frac{1 + \sum_{b=1}^{999} \mathbf{1}\{|T_b| \geq |T_{\mathrm{obs}}|\}}{1000},$$

with resolution 0.001. Benjamini–Hochberg correction (Benjamini and Hochberg, 1995) is performed within each of eight families: tissue (brain/PBMC) $\times$ confound specification (full/non-degree) $\times$ null (gene-label/row-shuffle). The full specification is the preregistered primary; the non-degree specification is reported alongside it as co-primary because the full specification conditions on proxy-derived degree covariates.

2.6 TF-disjoint supervised probe

The pooled audit is an unsupervised raw-readout test. As a supervised complement, run on paired PBMC multiome data only, we train a ridge probe to predict each TF's proxy-target profile from the FM graph itself. The probe was originally specified over per-gene embeddings; because the validated caches store gene-by-gene graphs rather than embeddings, we reformulated it at the pair level, which answers the same question without new model inference. Samples are ordered (TF, gene) pairs; features for each family are the edge weight $S_{tf,g}$, its reverse $S_{g,tf}$, and the within-TF rank of the edge. These are scalar per-edge summaries, so the probe tests edge-level linear recoverability, not row-level regulatory programs. Nothing in the design is indexed by TF identity, so a probe trained on 311 held-in TFs transfers unchanged to 135 disjoint test TFs. Regularization is chosen by leave-one-out cross-validation over training pairs. Performance is the per-test-TF Spearman correlation between predicted and observed proxy profiles, averaged over test TFs, after residualizing both sides against target peak count, gene length, GC content, and detection rate. The null permutes gene labels within each test TF; paired FM-versus-co-expression contrasts use sign-flipped per-TF differences. Gene-level degree features are withheld from the primary probe arm because they let the probe reach the proxy through gene-level confounding rather than through the graph's edges.

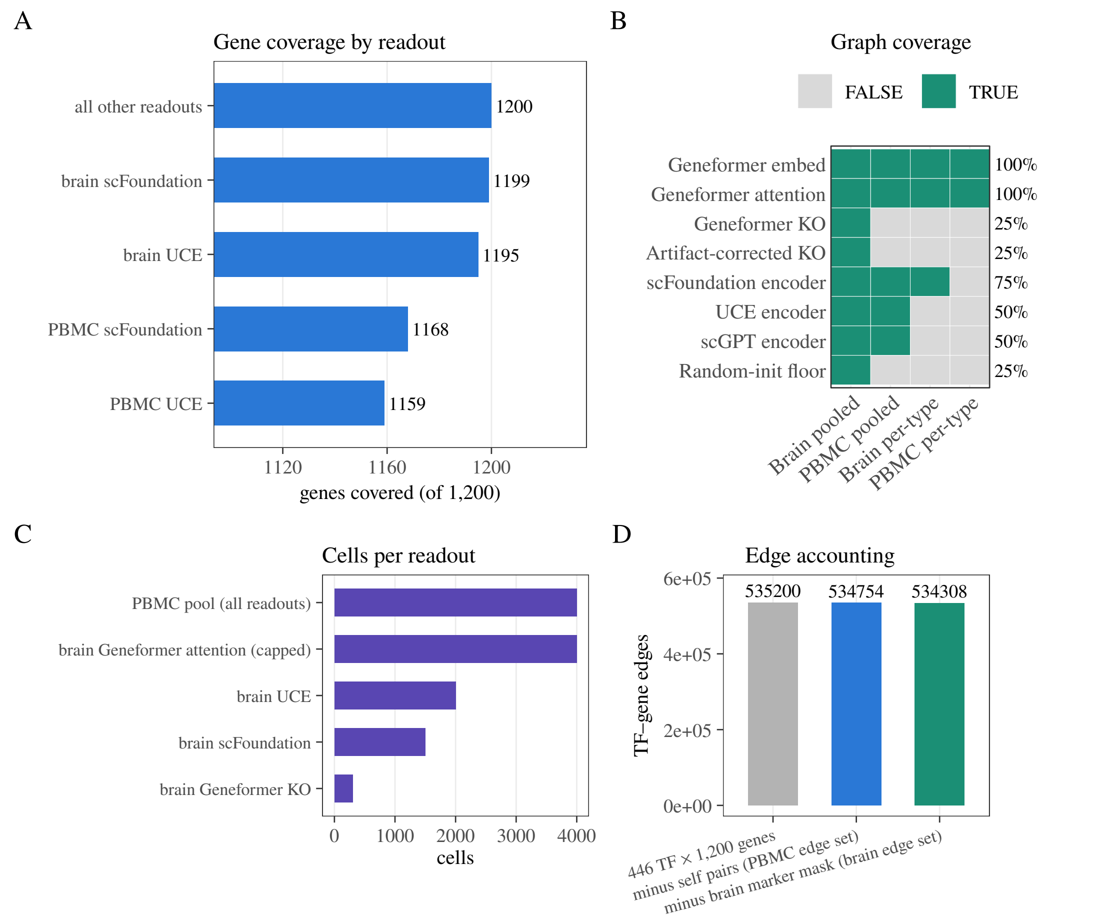

Figure 1. Coverage and fixed-panel quality control. (A) Manifest-gene coverage for encoder readouts. (B) Validated graph availability by tissue and analysis level; coverage is intentionally reported rather than assumed balanced. (C) Cell counts used by representative pooled inference runs. (D) Edge accounting from the $446 \times 1{,}200$ Cartesian panel through self-pair and brain marker-mask exclusions.

2.7 Descriptive and pipeline-sensitivity analyses

Per-cell-type rows report observed partial correlations, cell counts, ranges, medians, and sign counts. They have no randomization p-values or FDR adjustment. Cross-tissue proxy reproducibility is also observed-only. Finally, for each tissue we evaluate the following injected fractions:

$$\alpha \in \{0, .02, .05, .08, .12, .18, .25, .35, .5, .75, 1\}.$$

For each value, we combine an independently generated noise vector with an α fraction of the proxy residual and repeat the observed statistic 30 times. This axis-aligned diagnostic checks implementation response; it is not a power analysis and does not define an MDE, confidence interval, or exclusion bound.

The fixed panel, model/readout availability, inference cell counts, and edge-accounting steps are audited explicitly in Fig. 1; blank coverage cells denote missing validated graphs, not negative results. The full audit therefore comprises 13 pooled FM/readout rows rather than a balanced model-by-tissue factorial design.

3 RESULTS

3.1 The proxy is reproducible, but remains a proxy

The accessibility/motif construct shows moderate agreement across tissue consensus graphs: brain–PBMC $\rho = 0.5249$, brain–fibroblast-mix $\rho = 0.5245$, and PBMC–fibroblast-mix $\rho = 0.4623$

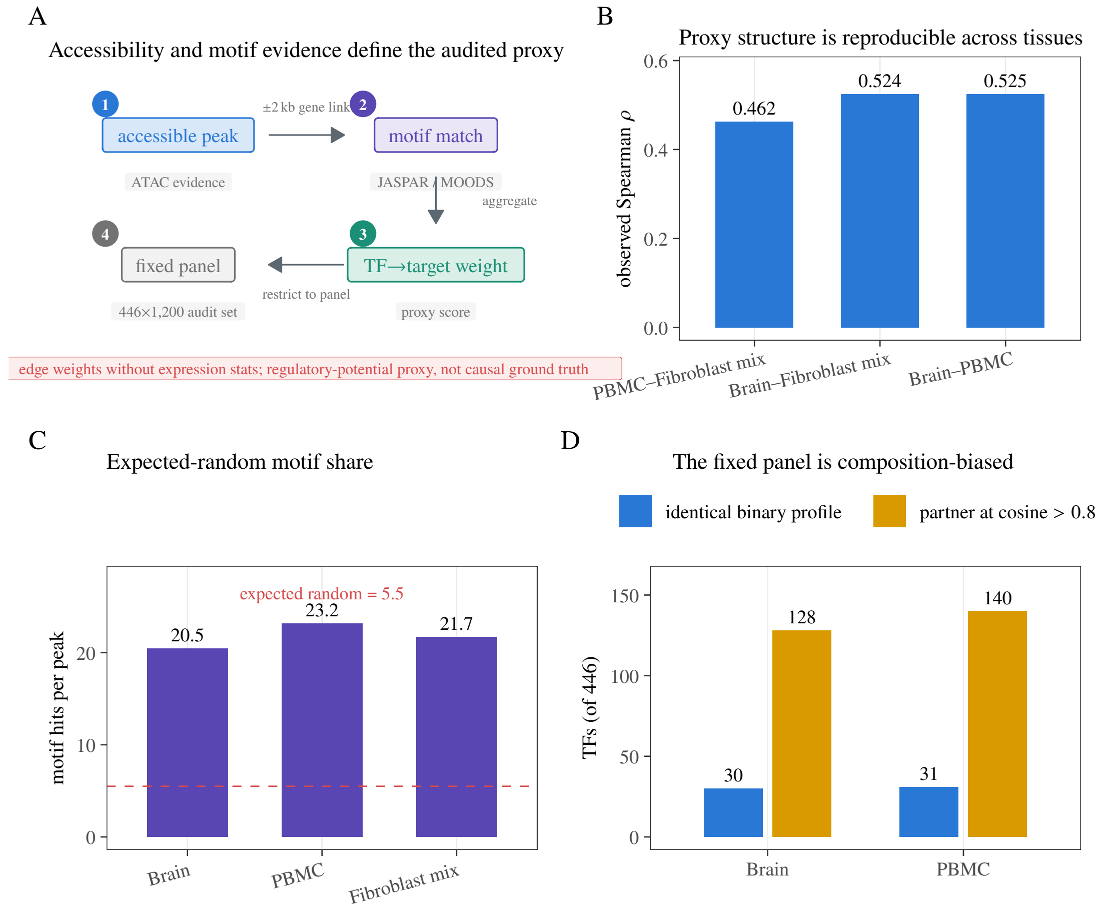

Figure 2. Regulatory-potential proxy, evidence, and panel composition. (A) Construction pipeline: accessible peaks are linked to genes (± 2 kb promoter-plus-gene-body), JASPAR/MOODS motif matches contribute directed TF$\rightarrow$target weights, and edges are restricted to the hash-pinned 446-TF$\times$1,200-gene panel; no RNA enters the proxy, which is not causal ground truth. (B) Observed consensus-graph correlations across three tissue datasets. (C) Motif hits per accessible peak relative to the expected 5.5 random hits at the 10^{-5} motif threshold. (D) TF target-profile redundancy: counts with an identical binary profile or a partner at cosine similarity > 0.8, out of 446 TFs.

(Fig. 2; Table 1). The third dataset (GSE206767) is a chemical-reprogramming experiment whose accessibility track pools day-0 fibroblasts and day-3 induced neuron-like cells, so it does not represent any resting tissue. These are descriptive fixed-panel correlations without new randomization inference. They show the construct is stable across ATAC inputs of very different provenance, not a causal ceiling or proof of TF–target truth. Figure 2 also makes the proxy's motif-noise and panel- composition limits explicit. Most of that stability is marginal structure: an additive TF-row-plus-gene-column model fit on one tissue's proxy predicts the other tissue's edges at $\rho = 0.32$–0.41 (69–78% of the observed agreement; Fig. 3), and after removing each tissue's own additive structure the residual agreement is $\rho = 0.41$–0.55. Table 1 reports the same quantities with edge Jaccard and mean per-TF-row Spearman. One pair deviates from that monotone reading: for brain–fibroblast-mix the additive fit is net-subtractive (residual $\rho = 0.55$ above the observed 0.52), so its additive-prediction fraction quantifies similarity in the marginals rather than decomposing the edge-level agreement. The dominant share of the apparent reproducibility is therefore fixed by the motif library and the proximity rule, both of which are tissue-independent by design, rather than by tissue-specific regulatory biology.

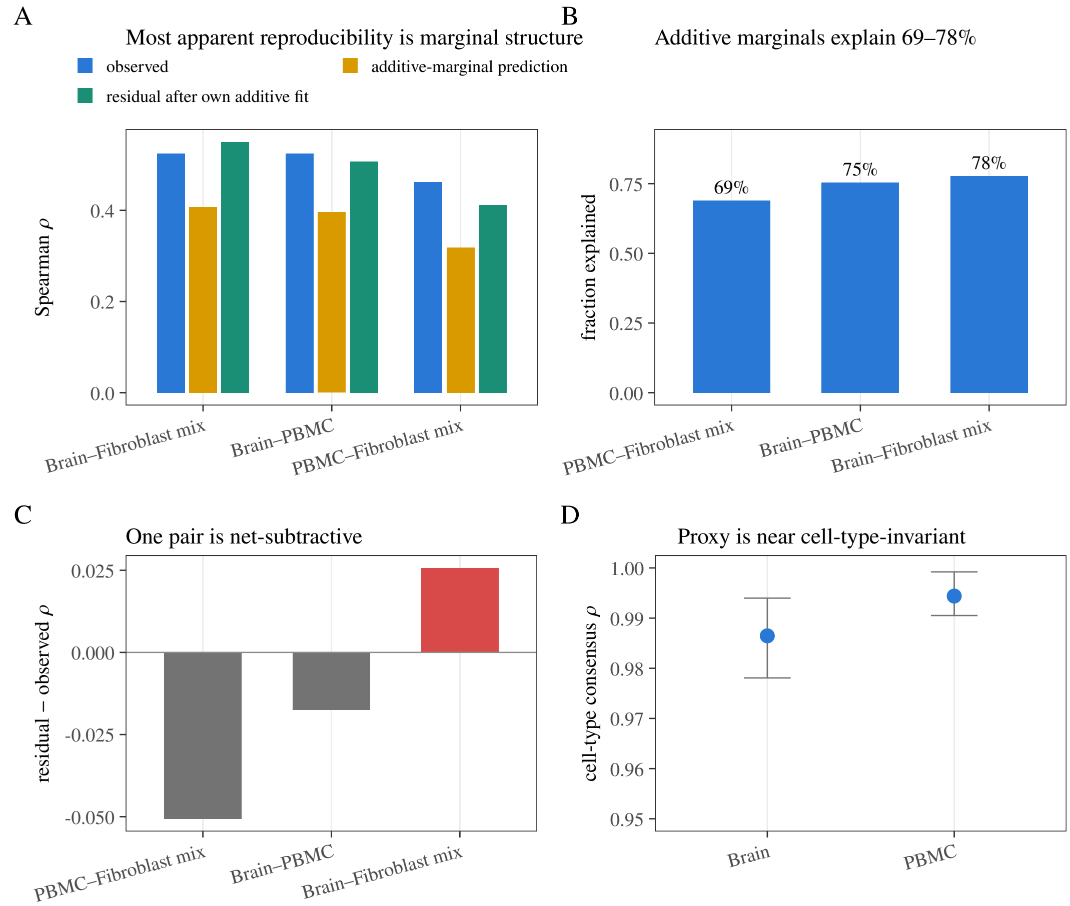

Figure 3. Cross-tissue structure of the regulatory-potential proxy. (A) Observed agreement, TF-row-plus-gene-column additive prediction, and agreement after removing each tissue's own additive fit. (B) Fraction of observed agreement predicted by additive marginals. (C) Difference between residual and observed correlation; brain–fibroblast mix is net-subtractive. (D) Range and mean of pairwise cell-type proxy correlations within brain and PBMC. These are descriptive fixed-panel quantities without new randomization inference.

Tissue pair	Obs. ρ	Add. frac.	Resid. ρ	φ	Edge Jac.	Mean row ρ	Shared TFs
Brain–PBMC	0.5249	0.7543	0.5073	0.5070	0.416	0.4001	446
Brain–Fibroblast mix	0.5245	0.7771	0.5502	0.5004	0.414	0.4151	446
PBMC–Fibroblast mix	0.4623	0.6895	0.4117	0.4461	0.374	0.3364	446

Table 1. Observed cross-tissue reproducibility of the regulatory-potential proxy on the fixed panel. Columns report observed Spearman ρ; the additive fraction of that agreement explained by TF-row plus gene-column marginals; residual Spearman after each tissue's own additive fit; binary support φ; edge Jaccard; mean per-TF-row Spearman; and the shared TF count. These are fixed-panel descriptive quantities with no confidence interval or new randomization inference.

3.2 Primary full-confound audit: six weak positives, one negative, and a baseline that matches them

Table 2 and Fig. 4 report all thirteen pooled FM/readout rows plus the co-expression baseline. Six small positive effects are supported by both randomization tests after within-family BH correction (Support column in Table 2). In brain, Geneformer embedding has $\rho = 0.00442$ ($q_M = 0.006$, $q_D = 0.003$), UCE encoder has $\rho = 0.00461$ ($q_M = 0.004$, $q_D = 0.003$), and scGPT encoder has $\rho = 0.01241$ ($q_M = 0.004$, $q_D = 0.003$). In paired PBMC, Geneformer embedding has $\rho = 0.00425$ ($q_M = 0.002$, $q_D = 0.003$), scGPT encoder has $\rho = 0.01098$ ($q_M = 0.002$, $q_D = 0.003$), and UCE encoder has $\rho = 0.00593$ ($q_M =$

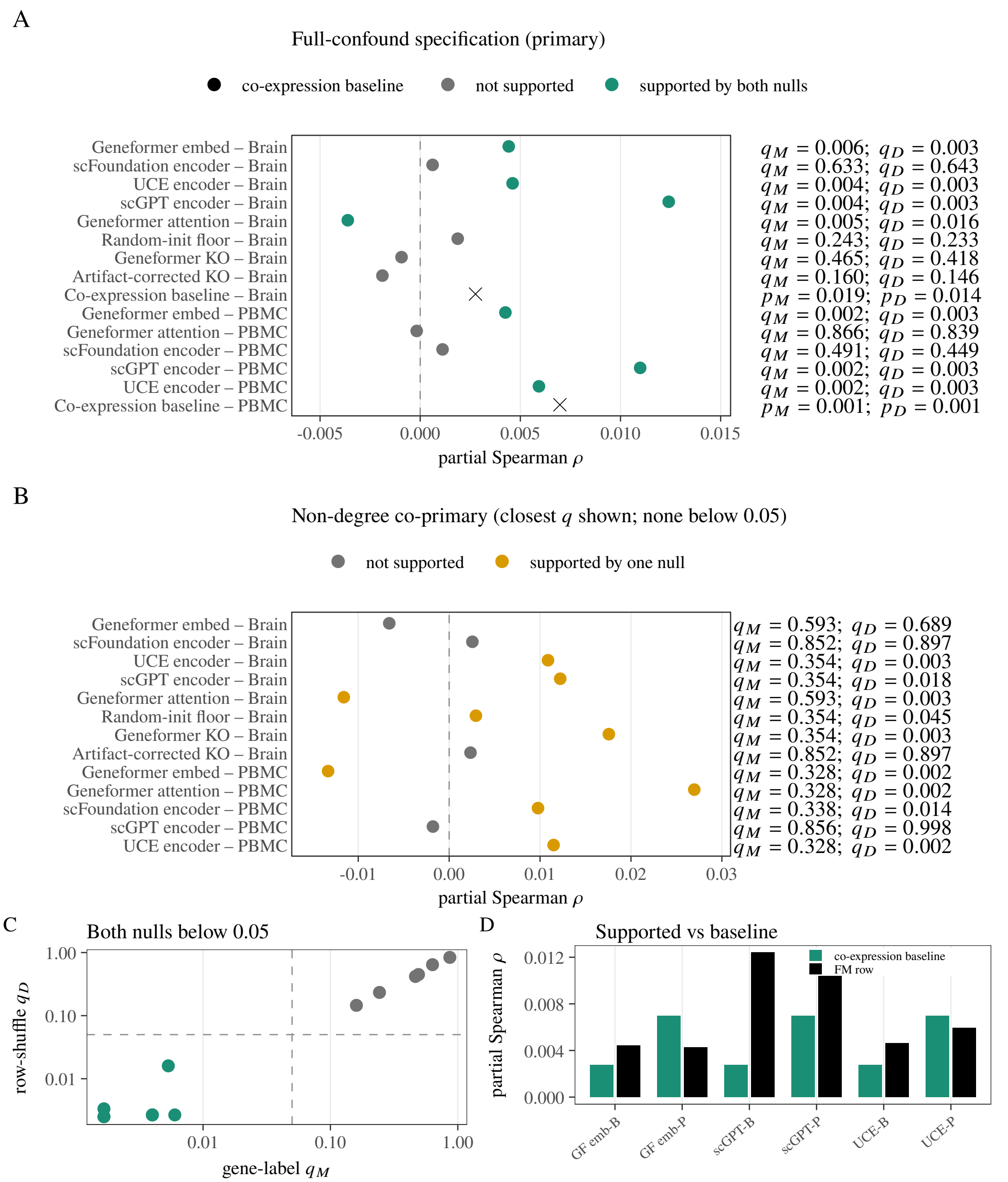

Figure 4. Primary fixed-panel audit under both prespecified confound specifications. (A) Full- confound pooled rows and the same-edge co-expression baseline; labels give BH-adjusted q_M and q_D for FM rows and uncorrected p_M, p_D for the baseline. (B) Non-degree co-primary rows; no row is supported by both nulls. (C) Full-confound q_M versus q_D; support requires both coordinates below 0.05. (D) Each supported positive row compared with its own-tissue co-expression baseline.

0.002, $q_D = 0.003$).

One further row is supported by both nulls with the opposite sign: brain Geneformer attention, $\rho = -0.00362$ ($q_M = 0.005$, $q_D = 0.016$), from the same model whose embedding readout is positive in the same tissue. A sign-inconsistent pair of readouts within one model is not what a regulatory-capability account predicts.

The remaining rows are not supported: brain scFoundation ($\rho = 0.00062$), Geneformer knockout (-0.00093) and its artifact-corrected knockout readout (-0.00189; the artifact-corrected row subtracts a

Tissue	Readout	Partial ρ	p_M	q_M	p_D	q_D	Support
Brain	Geneformer embed	0.00442	0.003	0.006	< 0.001	0.003	both
Brain	scFoundation encoder	0.00062	0.633	0.633	0.643	0.643	neither
Brain	UCE encoder	0.00461	< 0.001	0.004	< 0.001	0.003	both
Brain	scGPT encoder	0.01241	< 0.001	0.004	< 0.001	0.003	both
Brain	Geneformer attention	-0.00362	0.002	0.005	0.008	0.016	both (neg)
Brain	Random-init floor	0.00187	0.182	0.243	0.175	0.233	neither
Brain	Geneformer KO	-0.00093	0.407	0.465	0.366	0.418	neither
Brain	Artifact-corrected KO	-0.00189	0.100	0.160	0.091	0.146	neither
Brain	Co-expression baseline	0.0028	0.019	–	0.014	–	–
PBMC	Geneformer embed	0.00425	< 0.001	0.002	0.002	0.003	both
PBMC	Geneformer attention	-0.00018	0.866	0.866	0.839	0.839	neither
PBMC	scFoundation encoder	0.00112	0.393	0.491	0.359	0.449	neither
PBMC	scGPT encoder	0.01098	< 0.001	0.002	< 0.001	0.003	both
PBMC	UCE encoder	0.00593	< 0.001	0.002	< 0.001	0.003	both
PBMC	Co-expression baseline	0.0070	< 0.001	–	< 0.001	–	–

Table 2. Primary fixed-panel results. M denotes the gene-label null and D the within-TF-row degree-preserving target shuffle; each q is BH-adjusted within its tissue-by-specification-by-null family. Support is dual within-family BH threshold $q_M < 0.05$ and $q_D < 0.05$ ("both"); opposite-sign dual support is labeled "both (neg)"; "neither" otherwise. The co-expression baseline is tested on the same edge set outside the FM families and is marked "–" in Support; its entries are uncorrected p-values (dashes mark where no family correction applies).

random-token position-shift baseline and is not a positive control—the genuine positive control in this audit is the signal-injection diagnostic below), the random-init Geneformer floor (0.00187), and PBMC Geneformer attention (-0.00018) and scFoundation (0.00112).

The co-expression baseline, run through the same nulls on the same edge set (it cannot enter the FM families because partialling co-expression out of itself is degenerate), is itself supported in both tissues: brain $\rho = 0.00276$ ($p_M = 0.019$, $p_D = 0.014$) and PBMC $\rho = 0.00697$ ($p_M = 0.001$, $p_D = 0.001$). Two observations follow. First, the baseline reaches the same order of alignment as the supported FM rows. Second, the PBMC Geneformer-embedding positive ($\rho = 0.00425$) is smaller than its own baseline ($\rho = 0.00697$): on that row the model aligns with the proxy less than the confound it is adjusted against.

Four completed enhancement-v1 checks qualify the fixed-panel interpretation without changing the frozen primary families. Five-fold cross-fitting of the proxy-derived degree covariates preserved the qualitative direction in all 13 pooled rows, with absolute changes from the frozen estimates no larger than 1.1×10^{-5} (E1). Linkage sensitivity preserved the edge-weight structure for TSS–1 kb and TSS–5 kb policies ($\rho = 0.988$ and 0.971 versus the baseline), whereas a promoter-only policy changed the construct substantially ($\rho = 0.119$; E2). In 20 fixed 80% TF subsamples, signs were concordant in all brain and PBMC draws; the descriptive ρ ranges were 0.00271–0.00610 and 0.00115–0.00716, respectively (E3). Detection-based stratification also showed larger alignment for the operational housekeeping proxy than for low-detection TF strata (brain $\rho = 0.00891$ versus 0.00398; PBMC 0.02111 versus 0.00141; E5). These checks report observed stability and construct sensitivity; E3 did not recompute valid randomization nulls after TF subsampling, and E5 uses a detection-based operational label rather than a biological housekeeping annotation.

A separate sequence-opportunity audit (S1) applied a five-bin target-opportunity permutation and passed its null-calibration checks, but it is not the frozen v2 primary result: several row-level statistics and q-values differ under that stricter sensitivity design. We therefore report S1 as a sensitivity audit rather than claiming that the primary headline has been re-tested under S1. External binding calibration against ENCODE-compatible edges remains approval-blocked and is not treated as completed evidence. Our protocol pre-registers a readout *concordance* (attention-likeness) check: a co-expression-like readout should correlate positively with co-expression (attention $\approx$ co-expression is well established). This check blocks analogy to attention-GRN recovery stories; it does *not* adjudicate residual regulatory content on its own. The six dual-null positive rows fail concordance: their FM–co-expression Spearman correlations are -0.18 to $+0.01$ (all but one negative; PBMC scGPT is $+0.01$). The readouts that pass it

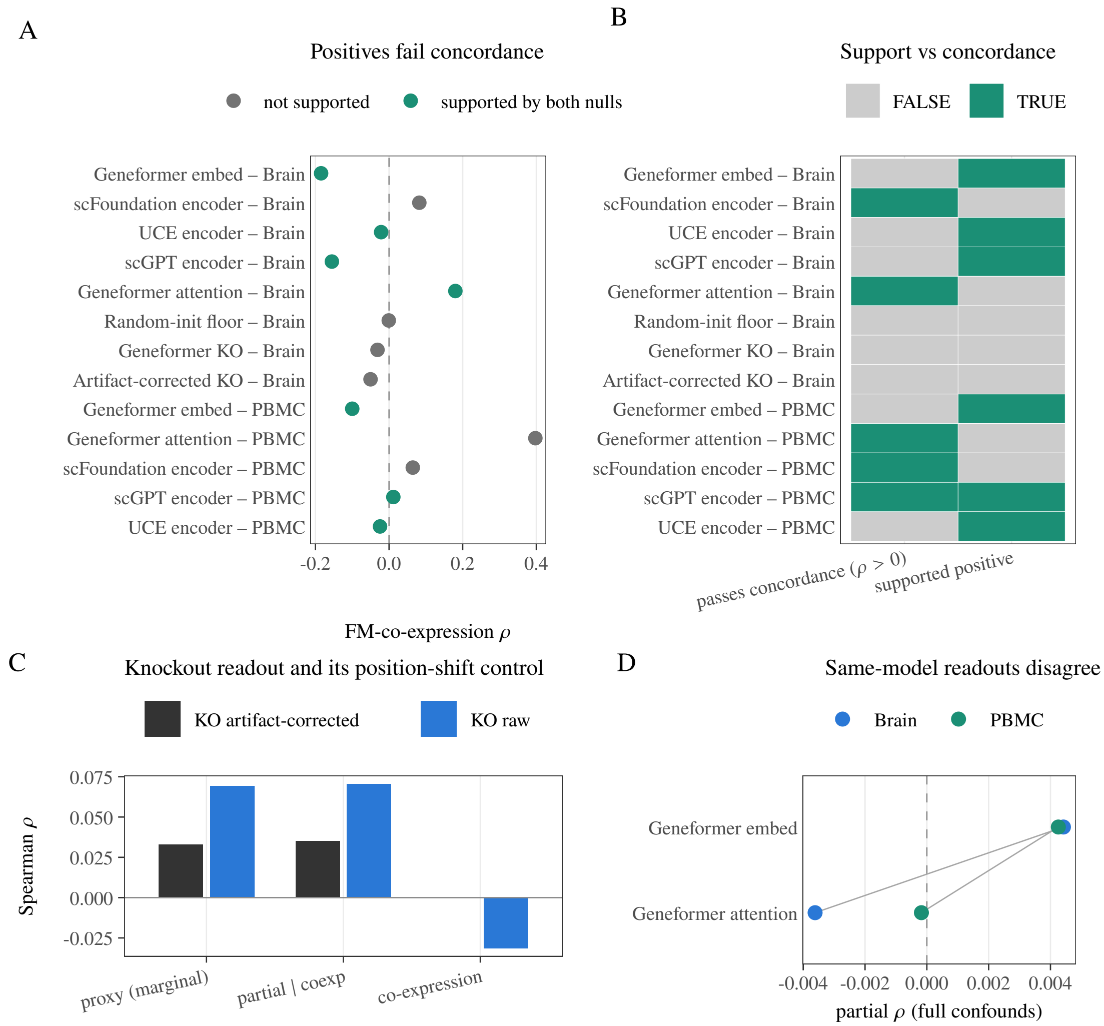

Figure 5. Readout concordance (attention-likeness) and internal consistency. (A) FM–co-expression correlation for all 13 pooled readouts, colored by full-confound dual-null support. (B) Positive joint-null support and positive concordance shown as separate binary checks; brain Geneformer attention is a concordance passer with negative dual-null support. (C) Marginal and partial behavior of the Geneformer knockout readout and its artifact-corrected knockout readout. (D) Geneformer embedding and attention effects within each tissue; the brain effects have opposite signs.

include scFoundation (+0.08) and PBMC Geneformer attention (+0.40); both are dual-null negative, so a concordance passer need not be dual-null supported (and conversely). Embedding-cosine anisotropy interacting with degree conditioning is a more parsimonious account of that inversion than regulatory encoding, so we report the six positives as readout-specific geometry rather than as regulatory signal (Fig. 5).

3.3 The alignments depend on conditioning on proxy-derived degree covariates

Removing the two degree covariates changes both effect direction and randomization behavior (Fig. 7). No non-degree row is supported by both nulls, and three of the six full-spec positives (brain Geneformer embedding, PBMC Geneformer embedding, PBMC scGPT) are negative before degree conditioning. Because the degree covariates are computed from the proxy graph itself, the full specification conditions on functions of the outcome; the positives therefore appear in exactly the specification that carries this hazard. We report the non-degree specification as co-primary and treat the divergence as a substantive finding, not a footnote: the row-shuffle null alone supports several non-degree rows while the gene-label null supports none, so degree adjustment and null choice define different finite-panel questions.

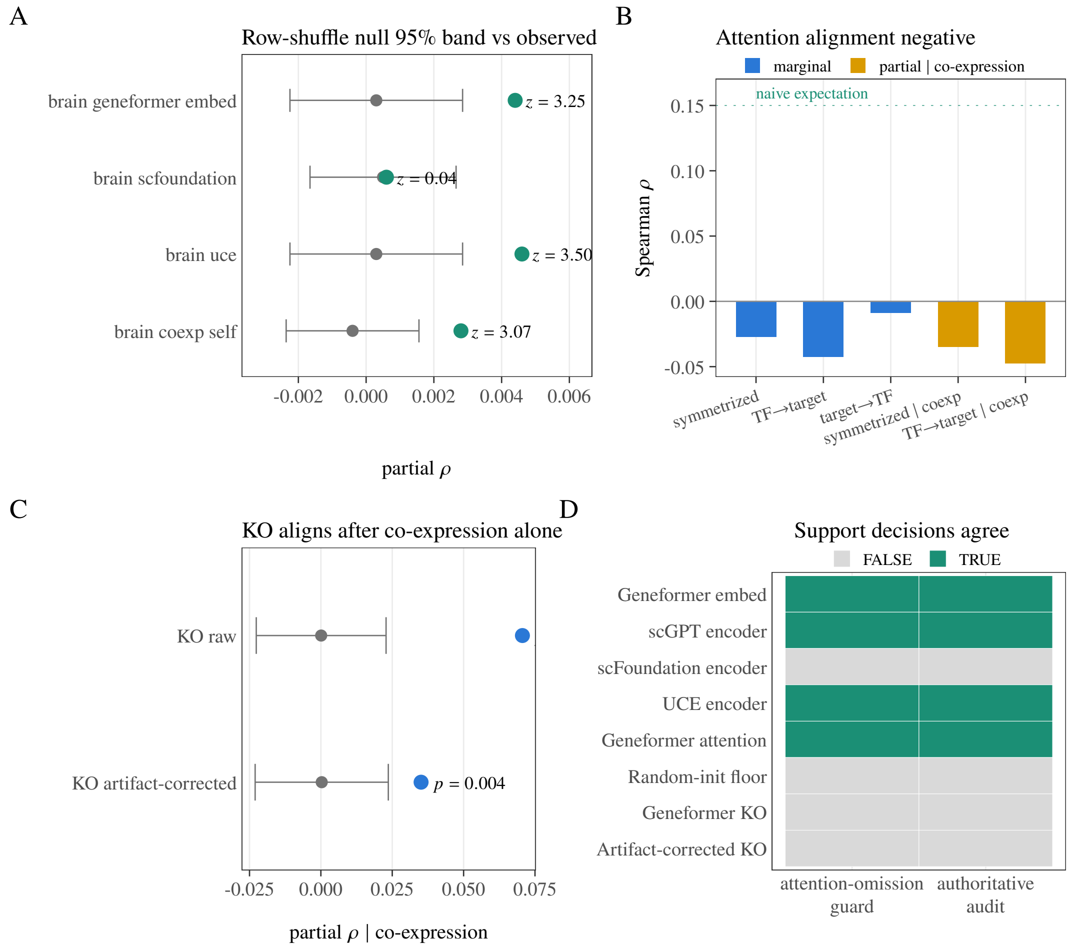

Figure 6. Randomization and readout diagnostics. (A) Observed statistics against the row-shuffle null mean and 95% null band for representative rows. (B) Brain Geneformer-attention marginal and co-expression-adjusted directions. (C) Geneformer knockout and artifact-corrected knockout both align with the proxy after partialling co-expression alone ($p = 0.001$ and $p = 0.004$); the audit's full confound specification removes that alignment (Table 2). (D) Joint-null support decisions are identical between the authoritative audit and the attention-omission guard.

3.4 Per-cell-type patterns are descriptive

The proxy itself is near cell-type-invariant on this panel (pairwise consensus $\rho \approx 0.99$), so per-cell-type rows score readouts against a reference with essentially no cell-type contrast; we report them only as descriptive robustness summaries. The full-confound cell-type analysis contains 15 brain (3 readouts $\times$ 5 cell types) and 14 PBMC (2 readouts $\times$ 7 cell types) FM rows per specification (Fig. 8). Brain full-confound estimates range from -0.00396 to 0.00445 with median 0.00258; PBMC estimates range from 0.00222 to 0.00682 with median 0.00413. Table 3 summarizes full and non-degree ranges. These rows have no randomization p-values or BH correction. They neither support cell-type discoveries nor provide population uncertainty.

3.5 The implementation responds monotonically to aligned injection, and the observed effects sit at the bottom of that ladder

The axis-aligned diagnostic rises monotonically from approximately zero at $\alpha = 0$ to mean partial correlations of 0.905 in brain and 0.913 in PBMC at $\alpha = 1$ (Fig. 9). The original ladder probes $\alpha \in \{0, .02, .05, .08, .12, .18, .25, .35, .5, .75, 1\}$; the later $\alpha \in \{0.002, 0.005, 0.01\}$ points are subdivided diagnostics used only to locate the observed magnitudes on that curve. They are not direct tests at each row's inferred equivalent and do not constitute a power analysis, MDE, confidence interval, or exclusion bound.

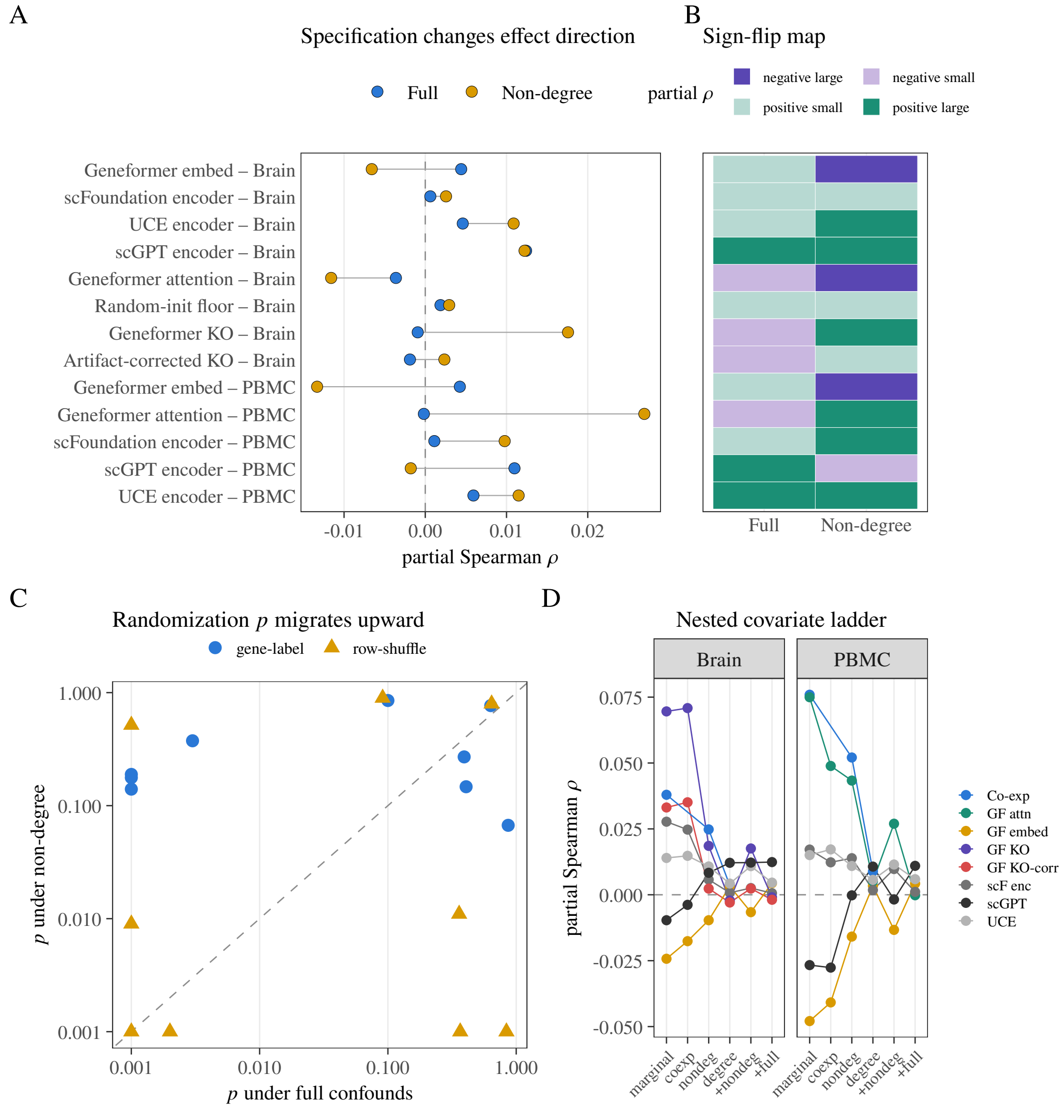

Figure 7. Dependence on the confound specification. (A) Full and non-degree estimates joined within each row. (B) Effect-sign map across specifications. (C) Full-confound versus non-degree randomization p-values for both nulls. (D) Nested covariate ladder from marginal correlation to the primary rung; lines are descriptive and are not a model-selection exercise.

Tissue	Confound spec	n_{rows}	ρ_{min}	ρ_{max}	ρ_{median}
Brain	full	15	−0.00396	0.00445	0.00258
Brain	non-degree	15	−0.00986	0.01625	−0.00053
PBMC	full	14	0.00222	0.00682	0.00413
PBMC	non-degree	14	−0.01303	0.05371	0.01502

Table 3. Per-cell-type descriptive ranges of observed partial ρ under full and non-degree confound specifications (brain: 15 exploratory FM×type rows; PBMC: 14). Min/max/median across those rows only; no randomization p-values, BH correction, or population-uncertainty interpretation.

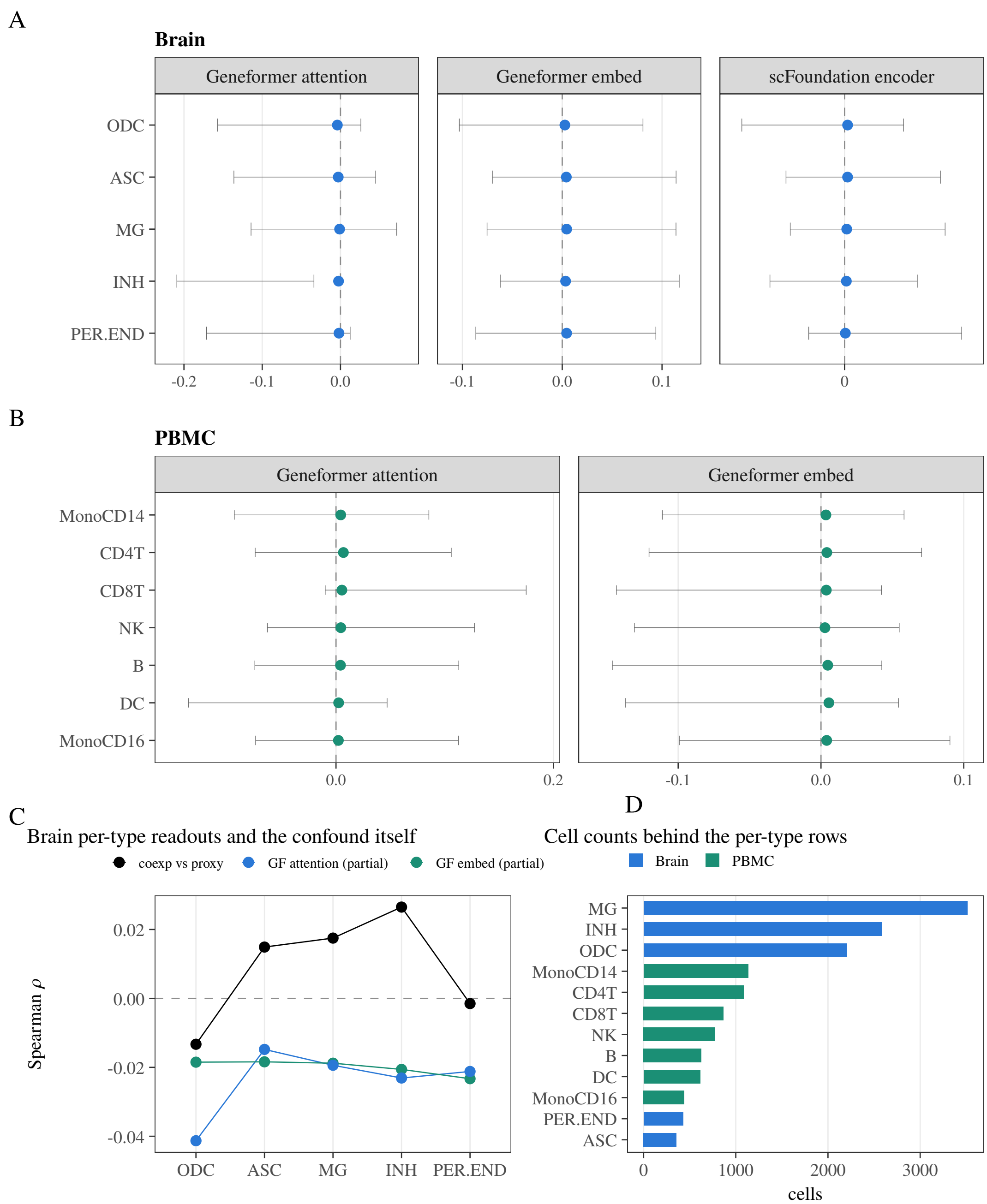

Figure 8. Per-cell-type descriptive robustness summaries. (A) Brain full-confound rows by readout. (B) PBMC full-confound rows by readout. (C) Earlier brain per-type co-expression, Geneformer- embedding, and Geneformer-attention summaries on the same fixed panel. (D) Cell counts underlying the per-type rows. These observed effects have no randomization p-values, multiplicity claims, or population-uncertainty interpretation.

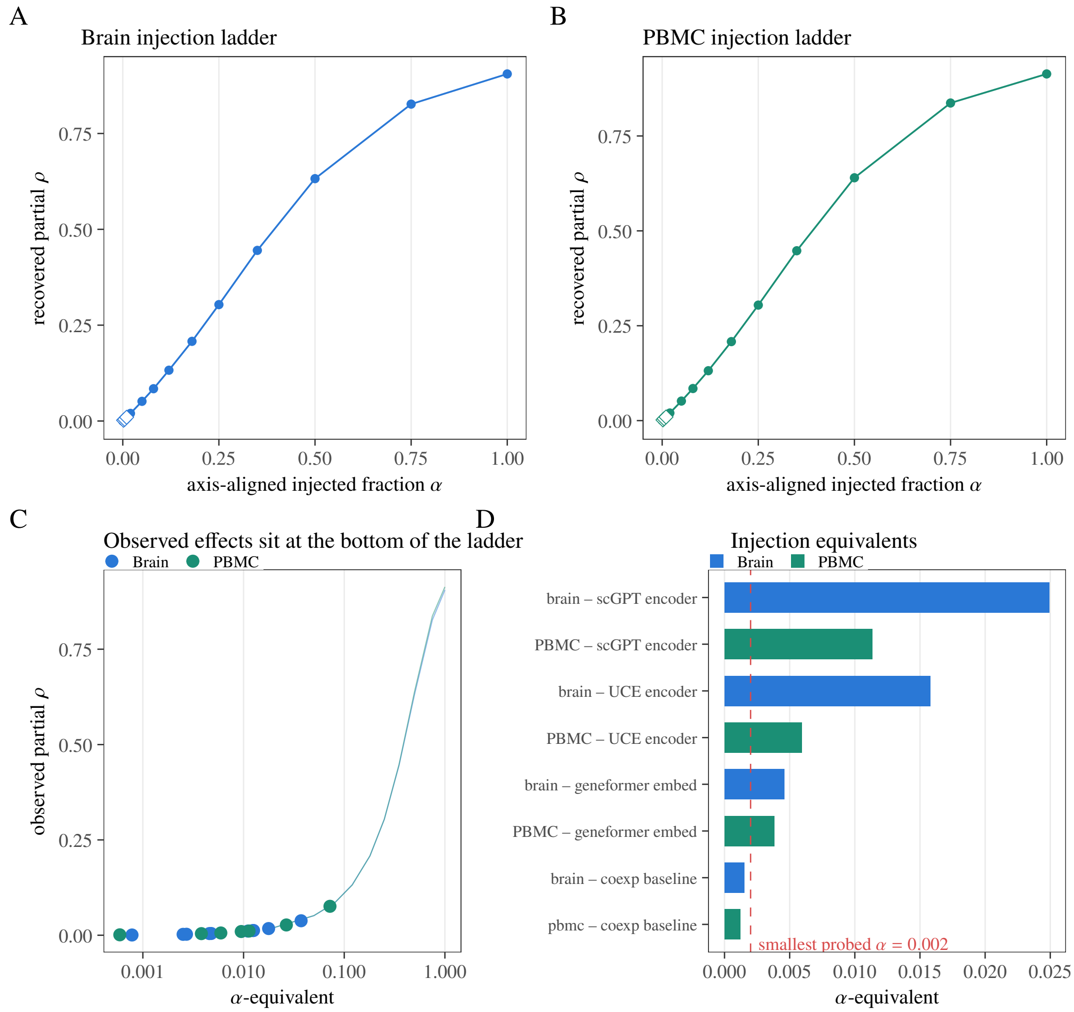

Figure 9. Axis-aligned pipeline-sensitivity diagnostic and observed-effect scale. (A–B) Brain and PBMC ladders; lines show the mean across 30 independent noise replicates, ribbons the replicate range, and diamonds the subdivided low-α points. (C) Observed effects located on the injection curves. (D) α-equivalents of the six supported positive rows relative to the smallest probed nonzero fraction; the negative brain attention row has no positive α-equivalent. This is not a confidence interval, power curve, MDE, or exclusion analysis.

3.6 A supervised TF-disjoint probe gives mixed family-level results

A linear probe trained to predict each TF's proxy targets from graph features on 311 held-in TFs and tested on 135 disjoint TFs (Fig. 10) gives mixed family-level results after confound adjustment. Paired sign-flip testing supports the UCE-versus-co-expression contrast ($q = 0.0325$) and the random-init floor contrast ($q = 0.0300$), while the Geneformer embedding ($q = 0.5125$), Geneformer attention ($q = 0.65$), and scGPT ($q = 0.255$) contrasts are not supported. Two properties of that test qualify the outcome. First, the Benjamini–Hochberg family contains the random-init floor—a methodological control rather than an FM readout—so that families are compared against the baseline on equal footing; excluding it ($n = 4$) raises the UCE contrast to $q = 0.052$, just above the 0.05 line. Second, the sign-flip statistic here compares how much each family is over-corrected, not how much it recovers: adjusted test correlations are essentially zero for UCE ($+0.0003$) and the floor ($+0.0023$) but negative for co-expression (-0.0120), so the supported contrasts mean the baseline is over-corrected further below zero, not that UCE carries more signal. The adjusted test correlations span -0.019 to $+0.002$ overall. The probe's confound set differs from the audit's—it contains neither co-expression nor the proxy-derived degree covariates—so it answers a different question (supervised recoverability, not raw-readout alignment) and neither confirms nor refutes the pooled rows. In particular, the random-init result prevents interpreting the UCE contrast

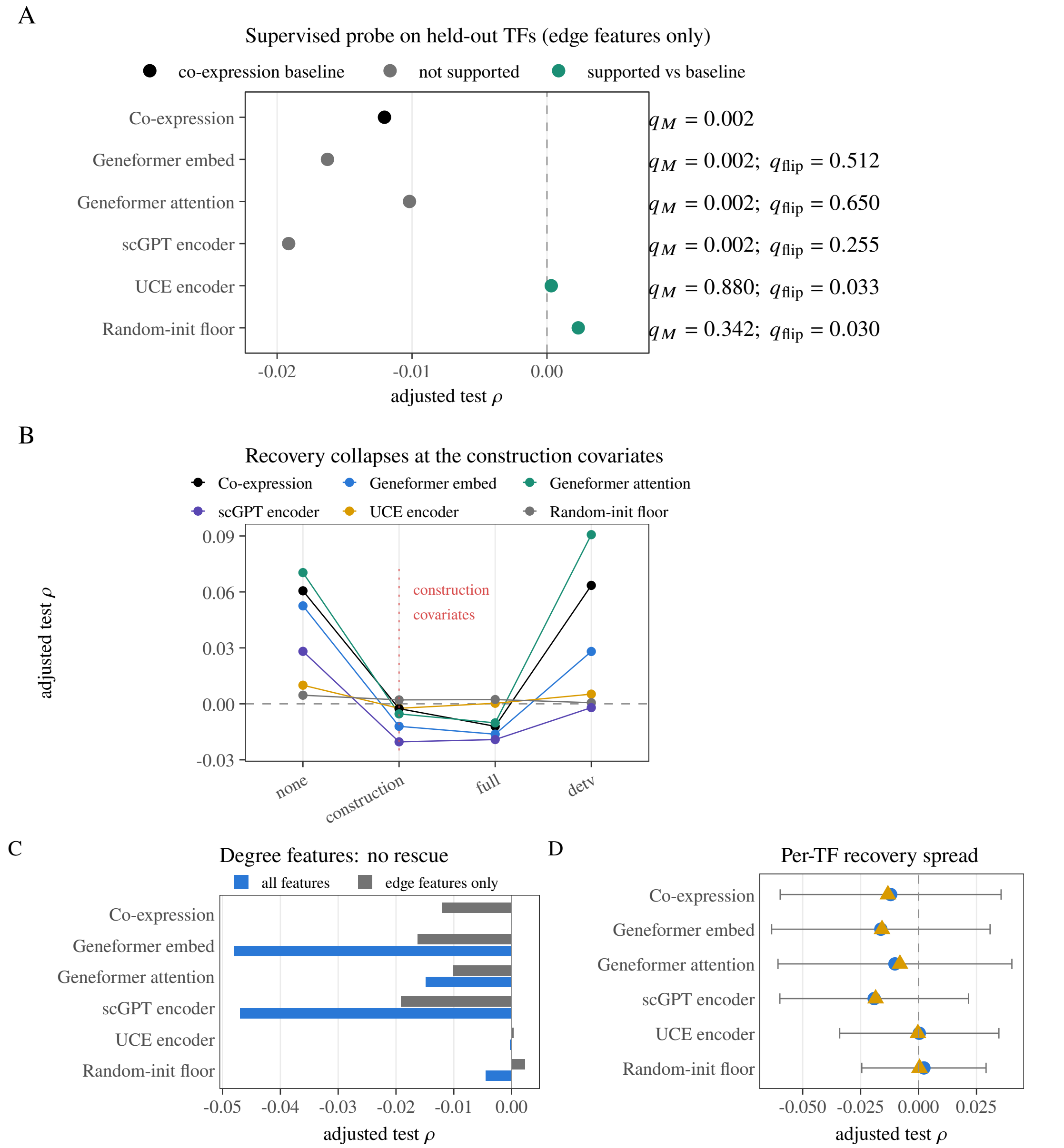

Figure 10. TF-disjoint supervised probe. (A) Confound-adjusted test correlation per family on 135 held-out TFs; labels give the gene-label permutation q_M and, for each FM family, paired sign-flip q_{flip} against co-expression. Green rows are supported against the baseline at $q_{\text{flip}} < 0.05$ (UCE 0.0325; random-init floor 0.0300). (B) Recovery across confound subsets. (C) Edge-only primary arm versus the all-feature sensitivity arm. (D) Across-TFmean, median, and one standard deviation of adjusted recovery.

as model-specific regulatory recovery, while the UCE result prevents a blanket claim that every FM is indistinguishable from co-expression.

3.7 The construct transfers across three independent ATAC datasets

The proxy was built independently in three datasets (brain GSE174367, PBMC 10x multiome, and the GSE206767 fibroblast reprogramming pool), and Fig. 11 quantifies how much of each dataset the construct actually uses and shares. The ± 2 kb linkage admits only a small fraction of each dataset's peaks: 8,823 of 219,070 in brain (4.03%), 6,609 of 111,743 in PBMC (5.91%), and 11,614 of 275,448 in the fibroblast pool (4.22%)—the proximity-linkage limitation is therefore a three-tissue property, not a PBMC-specific one. On the shared non-self TF–gene edge set (534,754 edges), 38,082 edges carry

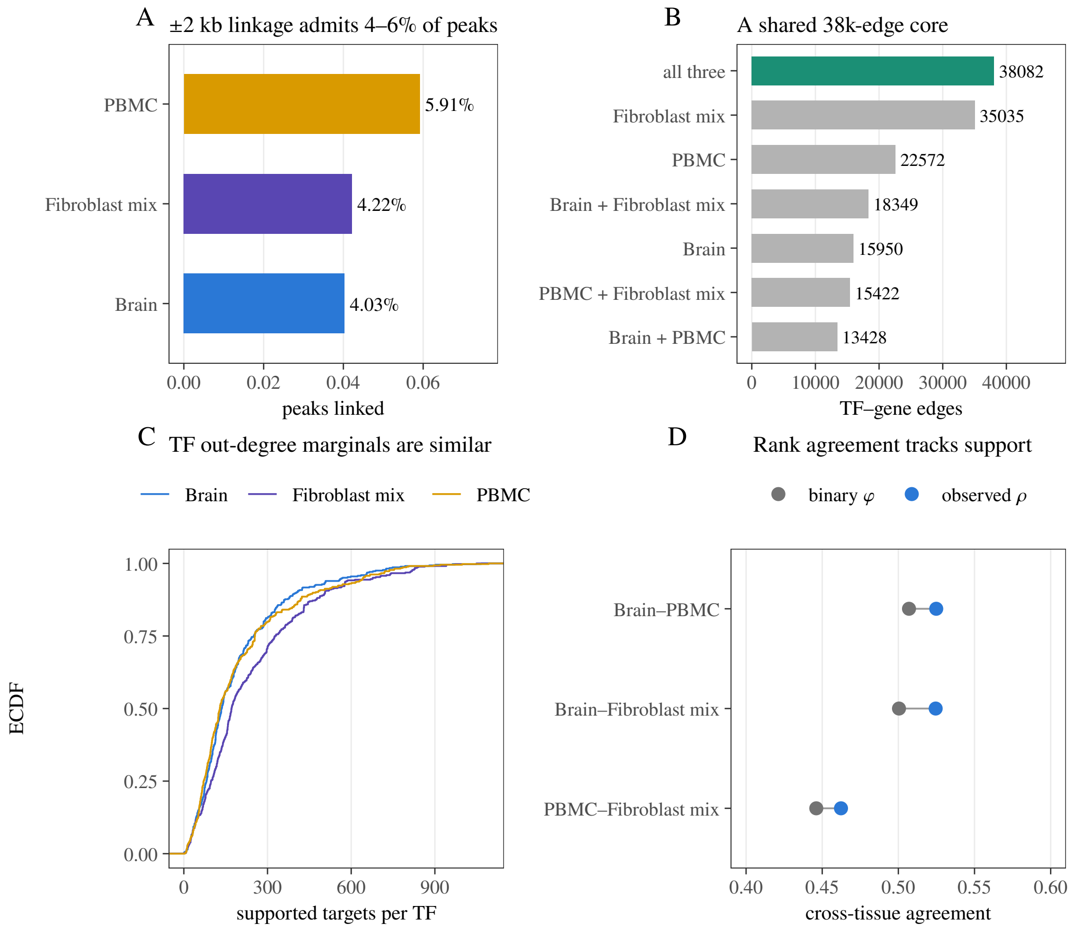

Figure 11. Third-tissue transfer of the proxy construct. (A) Fraction of each dataset's peaks admitted by the $\pm$2 kb linkage window (relevant over total peaks). (B) Overlap of supported (nonzero) non-self TF–gene edges across the three consensus proxies; the green bar is the 38,082-edge core present in all three. (C) Empirical CDF of TF out-degree (supported targets per TF) per tissue. (D) Observed cross-tissue Spearman ρ versus binary-support φ for the three tissue pairs.

nonzero weight in all three consensus proxies (24% of the 158,838-edge union), and the TF out-degree marginals are similar across tissues (median 129–171 supported targets per TF)—which is exactly the structure the additive decomposition of Fig. 3 attributes most cross-tissue agreement to. Cross-tissue rank agreement also tracks binary-support agreement (observed ρ 0.462–0.525 versus support φ 0.446–0.507), so the construct transfers chiefly at the level of which edges exist, not of their continuous weights. The fibroblast arm carries no foundation-model readouts and enters none of the audit's supported rows; it is a construct-level check, not evidence about model capability.

4 DISCUSSION

The audit finds six small positive fixed-panel alignments and one negative one, all supported by both randomization nulls under the full confound specification. Three features of that result restrain its interpretation. First, the co-expression baseline, tested on the same edge set, is itself supported in both tissues, and one of the six positives (PBMC Geneformer embedding) is smaller than its own baseline; the models do not separate from the confound they are adjusted against. Second, the two supported readouts of the same model in the same tissue (Geneformer embedding $+0.0044$, Geneformer attention -0.0036, both brain) have opposite signs, which is not the signature of a model-level regulatory capability. Third, three of the six positives are negative before conditioning on the proxy's own degree covariates, so they appear only in the specification that conditions on functions of the outcome. A supervised TF-disjoint probe, answering the distinct question of supervised recoverability on paired PBMC data (the probe does not cover the brain rows, whose RNA and ATAC come from unpaired sources), finds mixed contrasts: UCE and the random-init floor are supported against co-expression ($q = 0.0325$ and $q = 0.0300$), while Geneformer embedding, Geneformer attention, and scGPT are not ($q = 0.5125$, $q = 0.65$, and $q = 0.255$). The random-init result cautions against reading the UCE contrast as model-specific regulatory recovery.

The most defensible reading is therefore not "four models encode weak regulation." It is that small, readout-specific, specification-dependent alignments exist on this fixed panel, that they are concentrated in embedding-geometry readouts rather than attention readouts, and that under this audit they cannot be distinguished from residual confound structure (Sec. 2.1). A further observation points the same way: our own protocol pre-registers a concordance (attention-likeness) check in which a co-expression-like readout should correlate with co-expression (attention $\approx$ co-expression is well established). The supported positives come from readouts that fail that check (FM–co-expression Spearman -0.18 to $+0.01$), while concordance passers (scFoundation $+0.08$, PBMC Geneformer attention $+0.40$) are dual-null negative—so concordance does not adjudicate residual regulatory content, and dual-null Support without concordance is not recovery. Embedding-cosine anisotropy interacting with degree conditioning is a more parsimonious account of that inversion than regulatory encoding. This is compatible with reports that scFM-derived gene relationships often resemble co-expression (Kendiukhov, 2026b;a), and with zero-shot and biology-driven benchmarks that find scFMs unreliable or only weakly structured relative to simpler baselines (Kedzierska et al., 2025; Wu et al., 2025; Qiu et al., 2025). Failure to support a global regulatory-capability claim on this panel equally cannot support a field-level negative conclusion: the probe is linear and operates on pairwise graph summaries, the proxy is a narrow accessibility/motif construct, and other panels, tissues, or GRN-informed models could differ. Models pretrained with explicit GRN inductive biases (Yang et al., 2024; Kalfon et al., 2025; Hu et al., 2026) are outside the present multi-readout audit and remain a natural target for the same fixed-panel protocol.

Three statistical caveats apply throughout. First, one proxy randomization per replicate is shared across the rows of a family, so the Monte Carlo p-values within a family are positively correlated; Benjamini–Hochberg remains valid under positive regression dependence (Benjamini and Yekutieli, 2001), but the family-level error rate is then a joint property of the shared randomization—rejections co-occur across rows that share replicates, so the guarantee is weaker than the independent-per-row reading, not stronger. Second, in the non-degree specification the row-shuffle null preserves degree while the statistic does not condition on it, so the two nulls ask different questions there: several non-degree rows are unusual only under the row shuffle (brain Geneformer attention at $q_D = 0.003$ and the random-init floor at $q_D = 0.045$), which we read as evidence that degree adjustment—not regulatory content—drives that null's rejections. Third, replication depth differs by model: scGPT is supported in both tissues but not by the probe; UCE is supported in both tissues but its probe contrast depends on the family boundary; Geneformer is supported only through one of two readouts and its other readout is opposite-signed. We therefore report readout-level, not model-level, support.

Limitations and scope. The reference is a regulatory-potential proxy built from motif and accessibility evidence, and its coverage is narrow by construction. The ± 2 kb promoter-plus-gene-body rule retains 6,609 of 111,743 accessible peaks in the PBMC data (5.91%); distal enhancer elements, which carry most cell-type-specific TF occupancy (Thurman et al., 2012), are systematically excluded, and no co-accessibility or 3D-genome linker (for example Cicero (Pliner et al., 2018) or activity-by-contact models (Fulco et al., 2019)) is used. At the MOODS threshold used here ($p < 10^{-5}$, 500 bp accessible windows, both strands, 554 motifs), the expected random hit count is 5.5 per peak while the observed counts are 20.5 (brain), 23.2 (PBMC), and 21.7 (fibroblast mix), so roughly a quarter of motif hits are expected false positives (Wasserman and Sandelin, 2004); no information-content threshold, conservation filter, or PWM-length correction (Stormo, 2000) is applied. The panel is also composition-biased: 30 TFs in brain (31 in PBMC) share an identical binary target profile with at least one other TF (five FOX-family members are one such group), 128 of 446 TFs in brain (140 in PBMC) have a partner at binary cosine > 0.8, and in the reference lineage lists we checked, 7 of 21 neural TFs are present while all 37 housekeeping TFs are, because the panel cap ranks candidates by RNA detection rate. The proxy is near cell-type-invariant on this panel (pairwise $\rho \approx 0.99$), so per-cell-type rows score readouts against a reference with essentially no cell-type contrast. Brain analyses combine unpaired snATAC-seq with cross-study RNA, whereas the PBMC data are paired multiome, so the co-expression control is intrinsically stronger in PBMC than in brain. The full specification conditions on proxy-derived degree covariates; the non-degree co-primary specification avoids that hazard and supports no row under both nulls. The panel is fixed rather than sampled, so p_{MC} describes unusualness under the stated randomizations, not uncertainty over a TF or tissue population; no effect size carries an interval estimate, and with $N \approx 5.3 \times 10^5$ edges, effects of $|\rho| \approx 0.002$ are detectable, so statistical support alone cannot distinguish a small effect from no biologically meaningful effect. With $N = 999$ permutations, 0.001 is the resolution floor of every Monte Carlo p-value, not an exact value; rows with zero null exceedances are reported as $p < 0.001$. The supervised TF-disjoint probe is run on paired PBMC multiome data only; it does not directly test the eight brain rows, whose RNA and ATAC measurements come from unpaired sources. Per-cell-type and cross-tissue analyses are descriptive. The injection diagnostic is axis-aligned and cannot establish power against arbitrary biological alternatives. Fine-tuning, few-shot adaptation, decoder outputs, non-linear probes on raw per-gene embeddings, other gene panels, and direct perturbational validation are not tested. To make the path to broader validation explicit, we have inventoried and registered two additional single-cell RNA collections for follow-up replication: a development collection (27 datasets, $\approx 2.35 \times 10^6$ cells, mixed human and mouse) and a cancer collection (28 datasets, $\approx 1.20 \times 10^6$ cells, predominantly human), with per-file accession, cell count, and modality recorded in the project registry. These collections are RNA-only, so they cannot drive the accessibility/motif proxy used here; they are reserved for a planned co-expression-side replication that reuses the same fixed panel and confound specification once an orthogonal binding reference (for example ENCODE TF ChIP-seq) is in place.

5 CONCLUSION

scReg-Eval is first a fixed-panel audit protocol and software capsule: hash-pinned panel and nulls, dual-null Support reporting, concordance checks, and baseline comparison on a shared edge set. Applied to thirteen pooled model/readout rows against an accessibility/motif regulatory-potential proxy, it finds six small positive and one negative dual-null alignment under the full confound specification, a co-expression baseline that reaches the same order of alignment, sign-inconsistent readouts within the same model, and positives that depend on conditioning on outcome-derived covariates and that fail concordance. A supervised disjoint-TF probe gives mixed family-level contrasts, with the UCE and random-init floors distinguishing from co-expression but no model-specific regulatory recovery. These application-scene results neither support a global regulatory-capability claim for the audited graphs nor license a field-level negative conclusion beyond this protocol instance. The design complements living GRN benchmarks and standardized scFM task suites (Nourisa et al., 2025; Qiu et al., 2025) by building proxy edge weights without expression statistics while still disclosing panel and ATAC filters as expression-adjacent design choices.

DATA AND CODE AVAILABILITY

Code, fixed-panel manifests, seed metadata, model/readout coverage tables, authoritative audit outputs, and figure sources are available at `https://github.com/PeterPonyu/scfm-reg-audit` and archived at `https://doi.org/10.5281/zenodo.21724336`. The repository and archive support three levels of reproducibility: (i) validation that every published number and hash can be checked without external downloads; (ii) statistical reruns of the fixed-panel audit against the released outputs; and (iii) a full-pipeline recipe with dataset accessions, checkpoint hashes, a locked environment, and ordered commands. Public inputs include GSE174367 brain data (Morabito et al., 2021), the 10x Genomics 10k PBMC multiome (10x Genomics, 2021), and the GSE206767 chemical-reprogramming ATAC pool. Large upstream datasets and model weights are not redistributed; the availability record documents their stable source identifiers and the exact manifests used. GSE206767 is a chemical-reprogramming experiment: it contains day-0 BJ fibroblasts and day-3 induced neuron-like cells, pooled here as one accessibility track because no per-barcode condition metadata is distributed with the matrix (Gene Expression Omnibus, 2023).

ETHICS STATEMENT

This study reuses public, de-identified datasets and performs no new human-subject recruitment or intervention.

COMPETING INTERESTS

The author declares no competing interests.

FUNDING AND AUTHOR CONTRIBUTIONS

This work received no dedicated or external funding.

CRediT author contributions: Zeyu Fu: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Visualization, Writing—original draft, Writing—review and editing, and Project administration.

DECLARATION OF GENERATIVE AI USE

A large language model (Anthropic Claude, accessed through an agentic coding assistant) was used to assist engineering and refactoring of the analysis code, hardening of the statistical pipeline, copy-editing of the manuscript, and generation of the figure-generation and packaging scripts, including an internal simulated-review process used for quality control. No AI system generated the scientific data, the randomization outputs, or any reported number; every artifact was recomputed by the deterministic pipelines described in the Methods. The author reviewed and edited all AI-assisted outputs and takes full responsibility for the accuracy and integrity of the manuscript and the accompanying code. No AI tool is listed as an author. A disclosure package with representative before/after code and the prompts used is provided with the submission.

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